JeevanCare VitalWatch & Medicine Reminder Clock

Authoritative Hardware Schematic (CONNECTIONS.MD) | ESP32 DevKit Core Controller

Power: 5V DC Supply -> Power Switch -> ESP32 VIN. Regulated 3.3V LDO rail powers all sensors.

Shared I2C Bus: OLED (0x3C), MAX30102 (0x57), MAX30205 (0x4C), DS3231 RTC (0x68) on GPIO 19 (SDA) & GPIO 18 (SCL).

smart stick

File: smart_stick.kicad_sch

BT1 — 5V DC Supply / Li-ion

SW9 — Main Power Switch

PWR_FLAG

+5V

GND

U1 — ESP32 DevKit (NodeMCU)

VIN_5V

3V3

+3V3

R1 — 220R (5%)

D1 — Medicine Reminder LED

GPIO22

SW1 — Medicine ACK Button

GPIO21

SW2 — Screen / 2nd ACK Button

GPIO27

GPIO19_SDA

I2C_SDA

GPIO18_SCL

I2C_SCL

U2 — SSD1306 OLED 0.96in (0x3C)

U3 — MAX30102 PPG & SpO2 (0x57)

U4 — MAX30205 Clinical Body Temp (0x4C)

U5 — DS3231 Precision RTC (0x68)

VBAT

BT2 — CR2032 3V Backup

U6 — DHT11 (Temp & Humidity)

DATA

R2 — 4.7k Pull-up

GPIO4

U7 — AD8232 ECG Sensor

OUTPUT

LO+

LO-

3.3V

GPIO34

GPIO32

GPIO33

Master schematic generated from CONNECTIONS.MD

Sheet: /	
File: jeevancare.kicad_sch	
Title: JeevanCare VitalWatch & Medicine Reminder Clock	
Size: A3	Date: 2026-10-14
KiCad E.D.A. 10.0.7	Id: 1/2

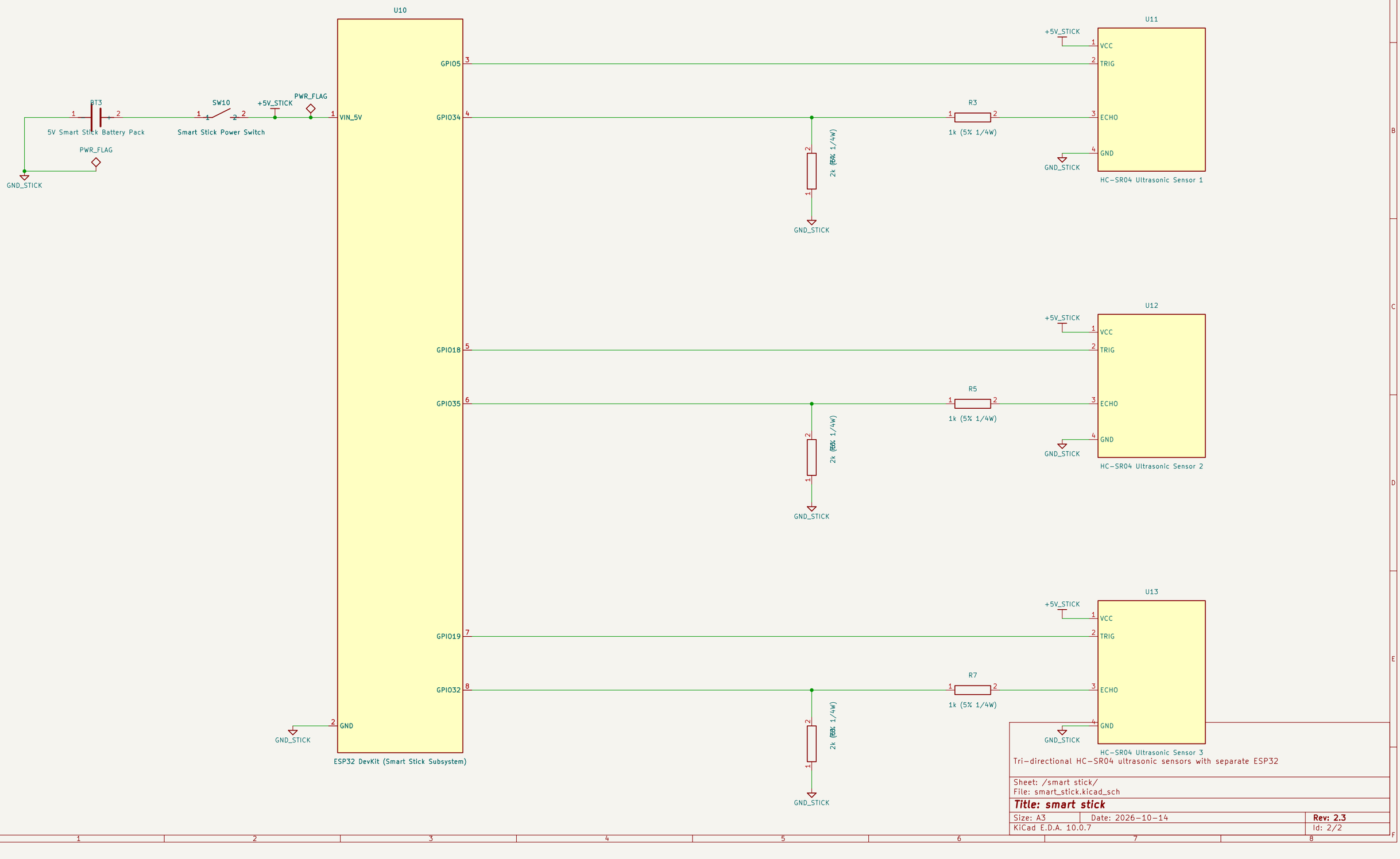
Rev: 2.3

Smart Stick Subsystem — Tri-Directional Obstacle Detection

Independent Embedded Controller: Dedicated ESP32 DevKit Unit for Smart Stick

Pinout: S1 TRIG=GPIO5 / ECHO=GPIO34 | S2 TRIG=GPIO18 / ECHO=GPIO35 | S3 TRIG=GPIO19 / ECHO=GPIO32

Logic Protection: HC-SR04 ECHO outputs 5V TTL. 1k/2k dividers safely step down signal to 3.33V for ESP32 inputs.



Tri-directional HC-SR04 ultrasonic sensors with separate ESP32

Sheet: /smart stick/		
File: smart_stick.kicad_sch		
Title: smart stick		
Size: A3	Date: 2026-10-14	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 2/2